

### Part I - Report for Data set 3

(a) Sample Mean = 63.6085

Sample Median = 63

Minimum = 28

Maximum = 98.5

Range = 70.5

(b) Sample Variance = 418.3934

Sample Standard Deviation = 20.45467

(c)

The sample mean, median, minimum, and maximum give important general statistics on the central tendency, spread, and extreme values of a dataset. The mean is 63.6085, which means that the average value of the dataset is around this value. The median is 63, which indicates the middle value of the sorted dataset. The minimum value is 28, which means the lowest score in the dataset, and the maximum value is 98.5, which is the highest score in the dataset. The range is 70.5, which provides the difference between the highest and lowest score in the dataset.

The sample variance and standard deviation show the amount in which the data points differ from the mean. The variance is 418.3934, which means a quite large amount of variability spread out from the mean in the dataset. The standard deviation is 20.45467, which provides a moderate spread of variability around the mean.

(d)

A graph is not only important to visualise relationships between variables in a dataset but also for understanding the context of the data being presented. That said, a histogram can be used to represent the data set given. A histogram can be used to represent the frequency distribution of the data in a dataset, thus allowing us to understand how the test scores are distributed. The distribution in a histogram can be evenly distributed, left skewed, right skewed or bell shaped. If the histogram is bell shaped or normally distributed, the mean, mode and median would be in the centre of the curve. With a histogram, we can find out which test score is the most common.

## Part II - Analysis and Findings of the Article

### Group

#### Question A

A total of 68 housewives, who fulfilled the inclusion criteria participated in the study. 18 were in the age-range of 21 to 40, 19 were from 41 years old to 50 and 26 were over the age of 50. Based on the data collected, the data type collected on the sociodemographic characteristics and cardiovascular risk factors of the housewives both qualitative and quantitative or more specifically, ordinal data type and ratio data type respectively. The ordinal data type can be observed in the education level of the housewives. Frequencies, median, mode, and percentages can be derived from ordinal data as observed. The ratio data type can be observed in age, duration of being a housewife, number of children, and household income which are under sociodemographic characteristics followed by BMI, WC, Systolic blood pressure, and diastolic blood pressure under cardiovascular risk factors. Ratio data is having a meaningful zero so they are unable to have negative values. This can be observed in the data collected under the variables stated. The statistical methods used in this study include descriptive statistics, data visualisation, ANOVA tests, Mann-Whitney U test, and the Kruskal-Wallis H test. Furthermore, negative binomial regression is used to determine the relationship between response variable, moderate-to-vigorous-physical-activity and the predictor variable. The mean and standard deviation differences between CVD risk factors, age group, and education level were calculated to be analysed. The mean differences between CVD risk factors and household income were also calculated. According to the study, the sample size used for this study is 68 participants according to the formulae by Cochran [25]: 
$$n = \frac{Z^2 \times p \times (1-p)}{e^2}$$
. In my opinion the sample size used in this study is not sufficient to draw a meaningful conclusion or result. The participants of this study are from low-cost residential area in Kuala Lumpur which aligns with the aim of the study. However, according to the law of large numbers, the larger the sample size, the closer the sample mean is to the population average. With the large number of residents living in low-cost residential area in Kuala Lumpur, a sample of 68 is not enough to provide relevant assessment of majority of the housewives in such regions. Granted, having a large sample size may not provide an accurate assessment of the population but at least a general overview can be provided.

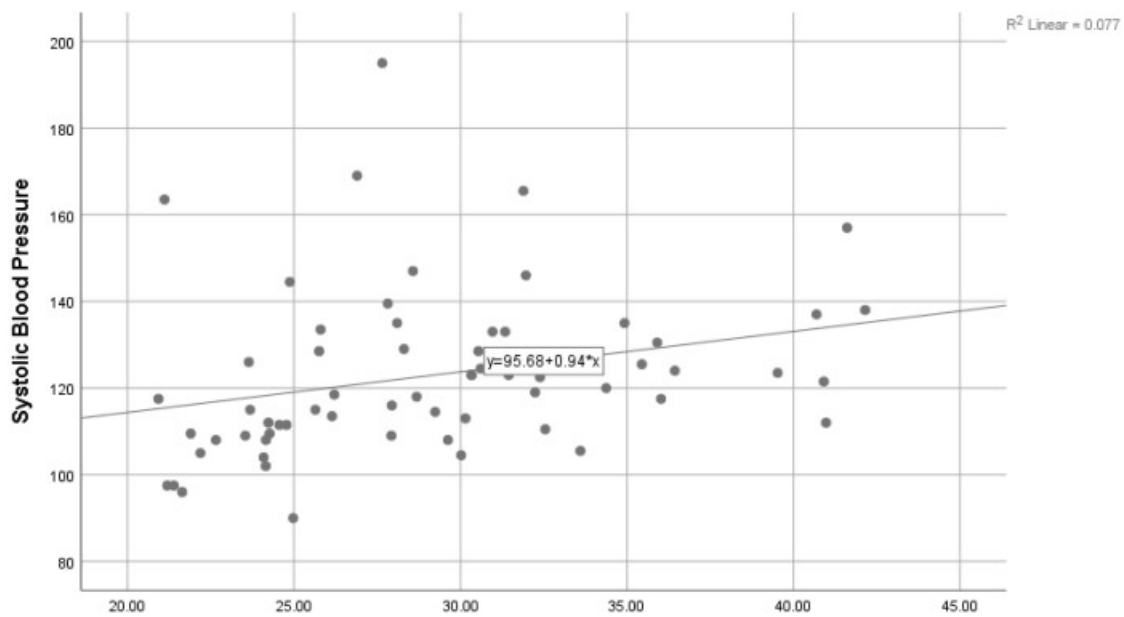
Article Links: <https://www.mdpi.com/1660-4601/18/11/6090>

## Question B

The mean difference between cardiovascular risk factors, age groups, and education level is calculated with a two-way ANOVA. By letting age group be the main factor, it is revealed that there was no significant mean difference between body mass index, waist circumference, diastolic blood pressure, fasting blood glucose, and total cholesterol. However, systolic blood pressure showed significant mean differences amongst age groups. This significant mean difference exists between the age groups 21-40 years and the 51-60 years according to a post-hoc test conducted. Moreover, if education level replaces age group as the main factor, it is discovered that there is no significant mean difference in all these cardiovascular risk factors. Upon further investigation, no significant mean differences between cardiovascular risk factors and family income exist. Among the 63 housewives, 57 had low physical activity, followed by 4 who showed moderate physical activity and 2 with a high level of physical activity. The study's  $451.35 \pm 617.78$  MET-min/week overall mean for total physical activity levels indicates that housewives engage in little physical exercise. The mean of moderate-to-vigorous physical activity (MVPA),  $70.48 \pm 282.36$  min/week, denotes physical inactivity because they did not reach the recommended 150 min per week.

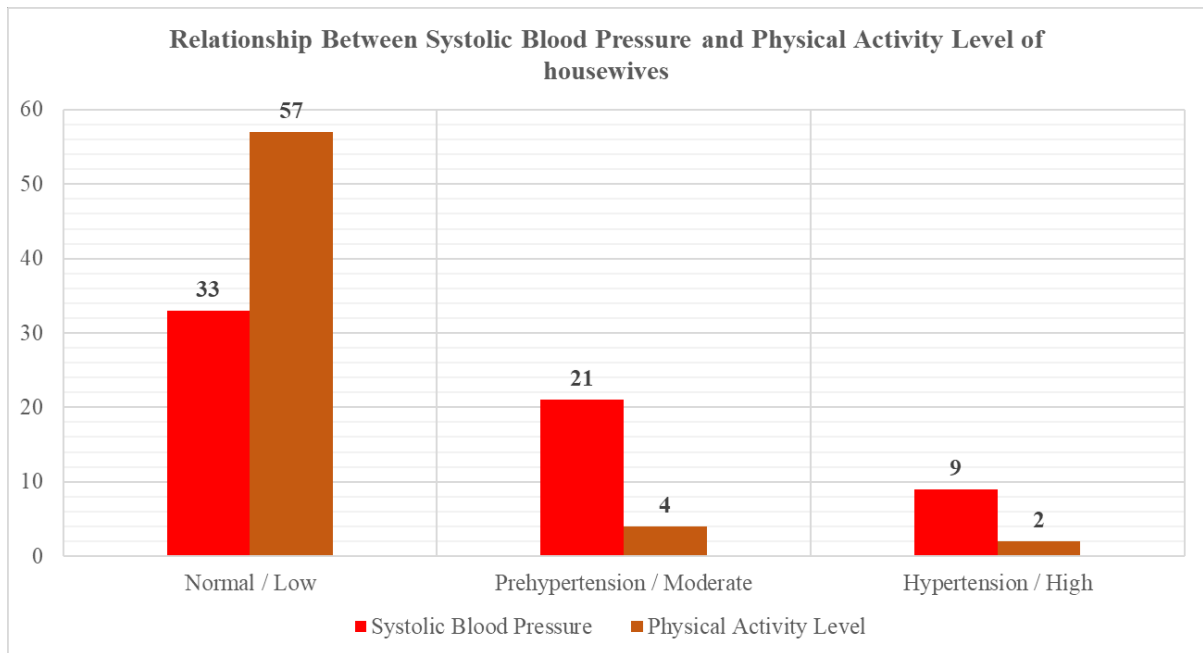
The authors concluded that housewives have higher a likelihood of being overweight or obese, which increases the chances of cardiovascular diseases based on the findings of this study. The prevalence rates for overweight and obesity in this sample, 28.6% and 58.7%, respectively, were found to be higher than those reported in the NHMS 2015 (Malaysia [24]) and the World Health Organisation [26], which had respective prevalence rates of 24.4% and 52.0%. However, the age range of adults in this study was 21 to 60 years old, but the NHMS 2019 studied adults aged 18 to 64 [23], which may have influenced the prevalence result. Furthermore, the participants of this study recorded a higher prevalence of abdominal obesity, which is measured by waist circumference [28], as compared to Spain with 33% [27] and Indonesia with 28%. Poor health is frequently associated with low socioeconomic status, which includes education, earnings, and employment. [4, 31-33]. The author backed this up by including an analysis of 882 Malaysians, which found that those without formal education and a salary less than RM 1000 are more likely to experience cardiovascular disease in the next ten years [32]. Nonetheless, the findings from this study revealed that there is no statistically significant relationship between family income and education level and cardiovascular risk factors. In this study, only a significant relationship between physical exercise and cardiovascular risk exists.

### Question C



**Figure 1.** Correlation between SBP (mm/Hg) and BMI (kg/m<sup>2</sup>).

There is a total of 7 tables and 2 graphs used in the article. The graph above is a scatter plot, which is about the correlation between Systolic Blood Pressure (SBP) and Body Mass Index (BMI). These scatter plots are effective visual aids in supporting the main points or arguments of the article since they display the correlation between blood pressure and body mass index. Scatter plots are extremely useful when displaying the pattern of how two variable changes together and determining if there is a linear relationship, clustering, or any outliers. Scatter plots are more effective than tables for presenting correlations because it helps analyse the linearity of correlations, visually convey the nuances of relationships, and make it possible to spot pattern and outliers. This dynamic visualization makes it easier for the reader to understand the complexity of the data and the relationship between SBP and BMI in the context of the article's key point. The systolic blood pressure and the body mass index of 63 respondents have been indicated above the scatter plot, which is the body mass index on the x-axis and the systolic blood pressure on the y-axis. A best-fit line with the formula,  $y = 0.94 * x + 95.68$  was provided on the scatter plot above.



**Graph 1.** Relationship Between Systolic Blood Pressure and Physical Activity Level of Housewives

The graph above is a bar graph, titled Relationship between Systolic Blood Pressure and the Physical Activity Level of Housewives. According to the bar graph above, out of the total 63 housewives, 33 have normal systolic blood pressure, 21 have prehypertension systolic blood pressure, and 9 have hypertension systolic blood pressure. Meanwhile, 57 out of 63 housewives have a low level of physical activity, 4 housewives have a moderate level of physical activity, and only 2 housewives achieve a high level of physical activity. From the graph above, we can see that only around half of the housewives have normal systolic blood pressure, and nearly 90% of the housewives have low levels of physical activity. Using the bar graph, we can easily identify the varying cardiovascular risk factors based on physical activity levels, by comparing the bar length representing each group. A vertically grouped bar graph is a strategic choice that complements the category character of the displayed data. The data provided on the graph above was divided into various categories; it is particularly effective at visually highlighting these categorical differences. Besides, the bar graph also summarises the data in simple visual form, and it allows viewers to easily identify the patterns or correlations within the dataset. Compared to a table, the bar graph provides a clear picture of data by presenting each data category in its frequency distribution.

## Individual

### Question D

#### Student A

The lack of geographical area in the study attenuates the impact of the research. A statistically non-significant result can be the consequence of an insufficient sample size – false negative error and the core study issue would not be adequately addressed by a sample size that was less than necessary. The research aims to determine whether a relationship exists between physical activity and cardiovascular risk factors amongst housewives in the low-cost residential area, but only in Kuala Lumpur. This limited geographical area will greatly affect the impact of the research. The lifestyle of the populace that lives in low-cost residential areas in different regions may differ slightly but can result in huge differences in the cardiovascular health of the populace.

The solution to this would be to expand the area of research. If the geographical area is expanded to different states or different regions in the same state, we can determine with greater accuracy if the relationship between physical activity and cardiovascular risk factors of housewives. For instance, the author can research in states such as Kedah, Perlis, Kelantan, and Terengganu to grasp the differences in lifestyles in different regions. By expanding the area of research, we inadvertently increase the sample size. A sample size that is larger will represent the population more accurately and hence provide a better result. Nonetheless, exceeding a certain point will only result in a slight increase in accuracy, thus wasting the extra effort and expenses.

## Student B

One of the limitations of this study is the location which this research is only conducted at high-rise housing areas. This is because housewives that live in high-level residential zones tend to be more active than those who live at ground-level residential zones. Housewives who live at tall buildings will have to walk more to reach their apartments even if the buildings have lifts. Sometimes, when the lifts are not working, they will have to climb the stairs leading to a higher level of physical activity. On the other hand, housewives who live at ground-level housing areas can easily reach their door in just a few steps. Hence, this disrupts the accuracy of the research as it is only focused on housewives who lived at high-rise housing areas as it is always expected to have a high level of physical activity. Thus, this cannot represent the physical activity of housewives as it is biased towards housewives who are generally more active.

There is a solution towards this limitation which is to include housewives from ground-level housing areas in the study too. If they are also involved in the research, this would make the results closer to the actual average physical activity of housewives which more realistic. This is because housewives from different housing areas have different lifestyles which influences their cardiovascular risks. All in all, including housewives from both high-rise and ground-level housing areas in the research can greatly improve the accuracy and the validity of the study.

## Student C

There is a potential weakness of study that found which is the measurement bias. Measurement bias, additionally referred to as information error, is when the data that was gathered for the study has errors or inaccuracies. This may occur for a number of reasons, including flawed assessment tools, biased interviewers, unclear respondents, or mistakes in the data input process. Misclassification of exposure and outcome variables as a result of measurement bias may undermine the study's internal validity and produce inaccurate estimates of associations. Researchers can improve the accuracy and validity of their findings by taking steps to reduce measurement bias during the study design and data collection phases. To make relevant and accurate findings on the relationship between physical activity and cardiovascular risk factors among low-income housewives in Kuala Lumpur, it is essential to precisely assess exposure and outcome variables.

To overcome the limitation, there are a several solutions that able to carry out. Researchers should make sure applying validated and reliable measurement instruments for assessing physical activity and cardiovascular risk factor before starting the survey to minimize the measurement bias in the end of result. By reducing the reliance on participants recall and avoiding the unequal information to the reality, kindly ask for the objective measures such as medical records, biometric data if feasible to complement self-reported data.



## Student D

One potential weakness of this study lies in the uncertainty surrounding the definition of physical activity and its relationship with household chores. Based on the article's publication date, we believe the data was collected from around late 2020 to 2021. During that period, Malaysia was experiencing a very severe Covid-19 pandemic, and most of the areas were under lockdown. Therefore, most of the outdoor activities have been cancelled, and many physical activities had reduced and transferred to online mode. During the lockdown, many housewives with children experienced an increase in workload, as they have to spend more on doing household chores and taking care of their children and spouse. The lack of a well-defined and universally accepted explanation of physical activity might lead to interpretation in participants, possibly distorting findings and weakening the study's reliability.

To address this issue and improve the research's credibility, a clear definition and categorization of physical activity is needed. It is critical to provide a precise and distinct description of physical activity particular to the research. This entails defining into several categories, including structured exercises such as aerobic workouts, and household tasks, and providing specific criteria to separate them. Besides, categorizing activities into groups will enhance the study's depth by clarifying the various contributions of different activities to whole physical activity levels.